

The Computer Vision Project

Today's Agenda

Introducing the Case

Going over the essential steps

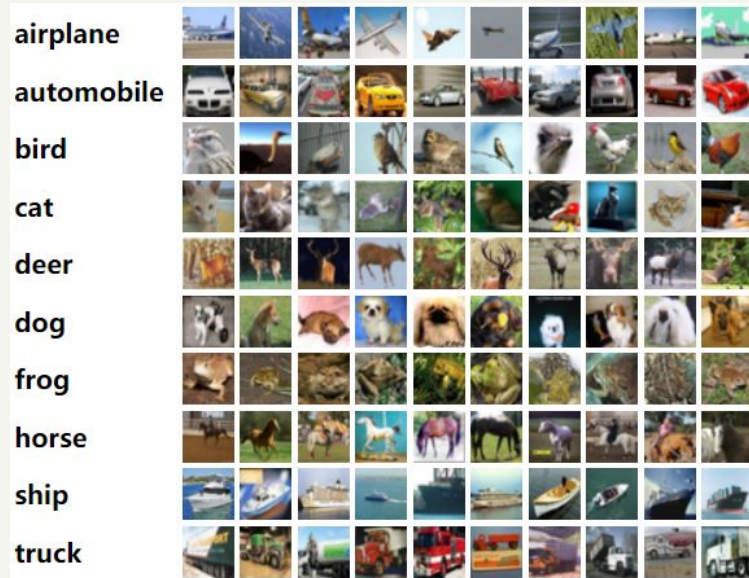
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Q&A

Introducing the Case

What is the project about?

- Classification problem with 10 classes to select from (CIFAR-10 Dataset)
- Colorful (RGB) images 32 by 32 in size.
- 50k images are available for training
- 10k images are available for testing



More about classes in the dataset

- The classes are completely mutually exclusive!

There is no overlap between automobiles and trucks. "Automobile" includes sedans, SUVs, things of that sort. "Truck" includes only big trucks. Neither includes pickup trucks.



Going Over Essential Steps

Availability of the Dataset

This dataset is also available in **TensorFlow & Keras**. Here is how you can import it:

```
from tensorflow.keras.datasets import cifar10

# Load CIFAR-10 dataset
(train_images, train_labels), (test_images, test_labels) = cifar10.load_data()
```

Limiting the dataset

As we've mentioned before there are:

- 50k images are available for training
- 10k images are available for testing

Using all 50k images may require additional computational resources as well as extra time for model training but will lead us to a greater performance!

To save time, let's limit training data to 10k only:

```
# Getting only 10k from the training portion of data  
n = 10000  
  
train_images = train_images[:n]  
train_labels = train_labels[:n]
```

Note 1: to achieve better performance all 50k images are needed

Note 2: testing data provides 10k images, so no need to limit it

Data Preprocessing & Model Selection

Key Steps to follow:

- 1. Remember that Neural Nets expect normalized images**

Your goal is to move pixel values from 1..255 to 0..1

Data Preprocessing & Model Selection

2. Import a pre trained neural network

ResNet50 might be a good candidate (EfficientNet is also an option). Initialize the model by setting the following parameters:

- **weights** - to import a neural network pretrained on ImageNet
- **include_top** - to drop off the top part of the neural network
- **input_shape** - to specify the input image size

Data Preprocessing & Model Selection

3. Build the top part of the neural net by:

- Including a layer that converts the feature maps into a single vector
- Introducing hidden layers

Our recommendation:

For the baseline, consider using a model with two hidden layers. The first hidden layer usually contains more neurons than the second. The number of neurons in each hidden layer is typically a power of two (e.g., 128, 64, 32, etc.).

Data Preprocessing & Model Selection

4. Finish up the top part of the neural net by including the layer where predictions are obtained

- Set the number of neurons in this layer to the number of classes you want your model to predict.
- Pick up the **activation function** for the multi class classification

Compile, Train, Evaluate

1. Compile your model by selecting:
 - An optimizer with the loss function, learning rate and metrics in it
2. Train the model by fitting it in and setting:
 - The number of epochs (**remember:** more epochs - more time!)
 - Validation data to validate the model on
3. Evaluate the performance on the testing portion of data to get unbiased view on how good the model is

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Let's move on to a notebook and explore the data!

Q&A